

1 Summary

Model (Run-name): model1 (test)

Iteration: 1

TAC: 116339.6843300442

Took: 0.32689666748046875s

Total Time: 0.32689666748046875s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

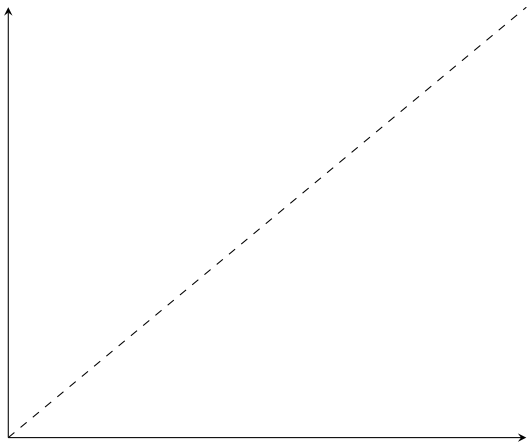
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

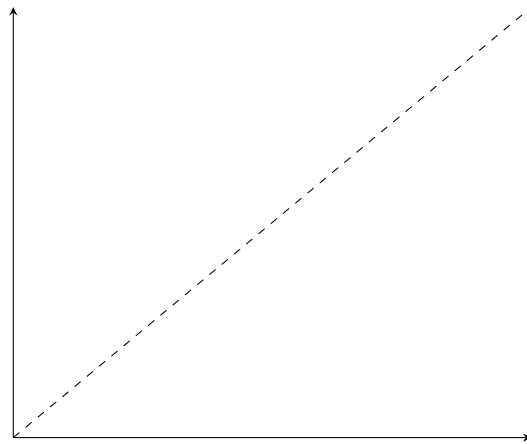
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.07471560670668927	$b_{1,1,1}^{H,out}$	5900.0
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.010093652445369401	$b_{1,2,2}^{H,out}$	3950.0
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.033101851851851855	$b_{2,1,2}^{H,out}$	4288.636363636365
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.011478725556538543	$b_{1,1,1}^{C,in}$	1025.0
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.009794883524078667	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.020865234510727387	$b_{2,1,2}^{C,in}$	4356.25
$A_{1,1,1}^\beta$	5.0	$q_{1,1,1}$	600.0	$b_{1,1,1}^{C,out}$	1625.0
$A_{1,2,2}^\beta$	39.36524453694061	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	28.333333333333332	$q_{2,1,2}$	2550.0	$b_{2,1,2}^{C,out}$	6906.25
$A_{cu,1}^\beta$	2.9819181007907165	$q_{cu,1}$	250.0		
$A_{cu,2}^\beta$	24.97364426355679	$q_{cu,2}$	1850.0		
$A_{hu,1}^\beta$	4.943755299006499	$q_{hu,1}$	450.0		
$A_{1,1,1}$	5.0	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	39.36524453694061	$t_{1,2}^{(H)}$	590.0		
$A_{2,1,2}$	28.333333333333332	$t_{1,3}^{(H)}$	395.0		
$A_{cu,1}$	2.9819181007907165	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	24.97364426355679	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	4.943755299006499	$t_{2,3}^{(H)}$	462.5		
$dt_{1,1,1}$	15.83882903021417	$t_{1,1}^{(C)}$	620.0		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	580.0		
$dt_{1,2,2}$	90.0	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	45.0	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	10.0	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	24.14596012830328	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	75.0	$t_{1,1,1}^H$	590.0		
$dt_{cu,2}$	142.5	$t_{1,2,2}^H$	395.0		
$dt_{hu,1}$	60.0	$t_{2,1,2}^H$	370.0		
$f_{1,1,1}^H$	10.0	$t_{1,1,1}^C$	450.0		
$f_{1,2,2}^H$	10.0	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	11.590909090909093	$t_{2,1,2}^C$	580.0		
$f_{1,1,1}^C$	2.5	$b_{1,1,1}^{H,in}$	6500.0		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5900.0		
$f_{2,1,2}^C$	10.625000000000002	$b_{2,1,2}^{H,in}$	6838.636363636366		

3 *RecLMTD* Tangent Points

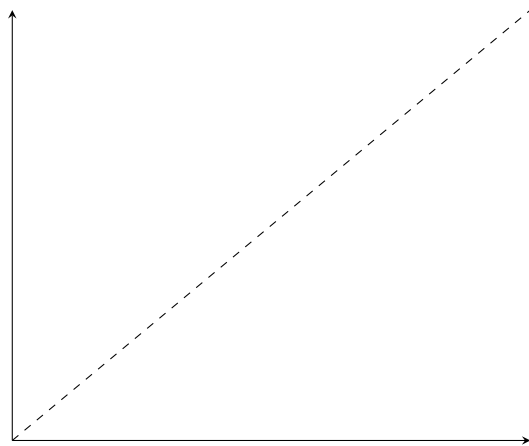
- $(1, 1, 1)$
 - Value: 0.074716
 - Error: 0.004046652544216
 - Relative error: 0.051378065874482
- $(1, 2, 2)$
 - Value: 0.010094
 - Error: 0.005309618233740
 - Relative error: 0.344707195267396
- $(2, 1, 2)$
 - Value: 0.033102
 - Error: 0.029215020431393
 - Relative error: 0.468813972219335
- $cu, 1$
 - Value: 0.011479
 - Error: 0.002319848740852
 - Relative error: 0.168122350240959
- $cu, 2$
 - Value: 0.009795
 - Error: 0.000009899340188
 - Relative error: 0.001009643999793
- $hu, 1$
 - Value: 0.020865
 - Error: 0.002239671507937
 - Relative error: 0.096934889331650



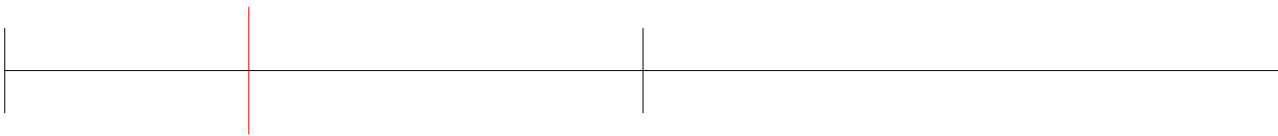
(a) *RecLMTD* - (1, 1, 1)



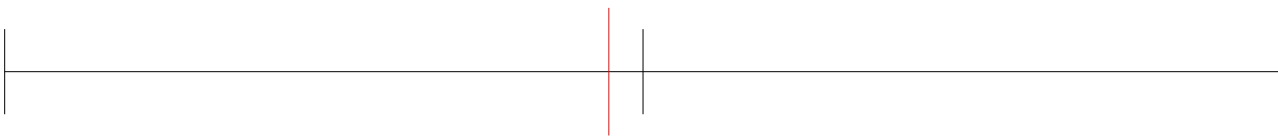
(b) *RecLMTD* - (1, 2, 2)



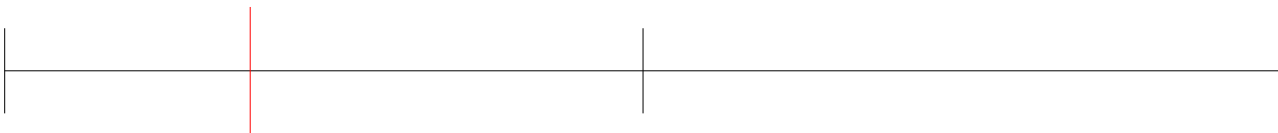
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - *cu*, 1



(b) *RecLMTD* - *cu*, 2



(a) *RecLMTD* - *hu*, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5900.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 6838.636363636366
 - Absolute error: 9.094947017729282e-13
 - Relative error: 1.3299357553342916e-16

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 5900.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 3950.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 4288.636363636365
 - Absolute error: 0.0
 - Relative error: 0.0

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 1025.0
 - Absolute error: 425.0
 - Relative error: -0.29310344827586204
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 4356.25
 - Absolute error: 9.094947017729282e-13
 - Relative error: -2.087792715691083e-16

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 1625.0
 - Absolute error: 500.0
 - Relative error: 0.4444444444444444
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 6906.25
 - Absolute error: 743.7499999999991
 - Relative error: 0.12068965517241363



(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



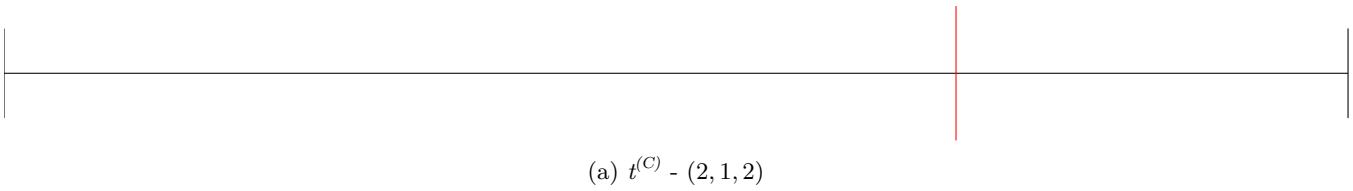
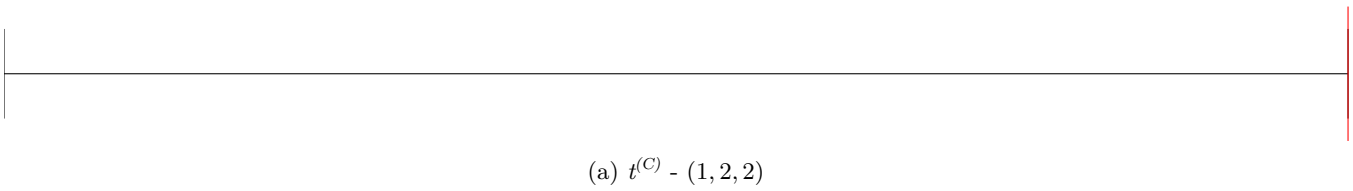
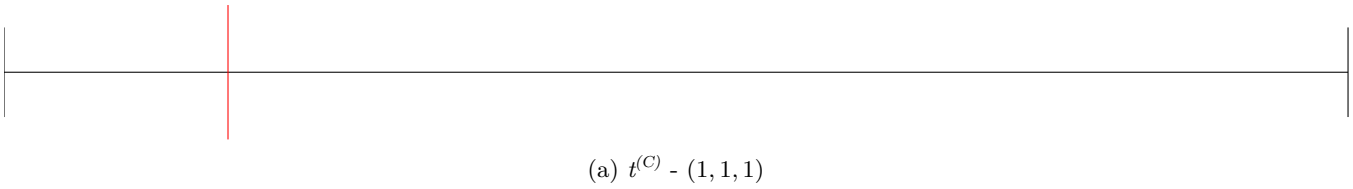
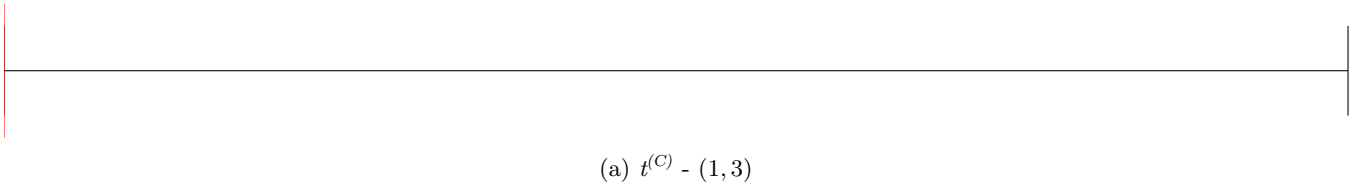
(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$



5 Area Breakpoints

5.1 A

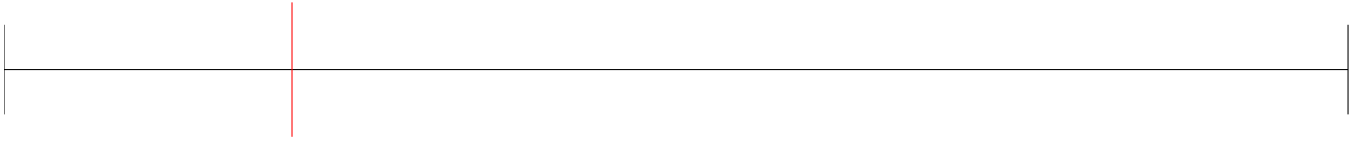
- $(1, 1, 1)$
 - Value: 5.000000
 - Error: 84.658728048027115
 - Relative error: 0.944232980894825
- $(1, 2, 2)$
 - Value: 39.365245
 - Error: 0.0000000000000057
 - Relative error: 0.000000000000001
- $(2, 1, 2)$
 - Value: 28.333333
 - Error: 140.48611111111114
 - Relative error: 0.832167832167832

5.2 A_{CU}

- $cu, 1$
 - Value: 2.981918
 - Error: 2.757444677478555
 - Relative error: 0.480444394962967
- $cu, 2$
 - Value: 24.973644
 - Error: 11.267424775534277
 - Relative error: 0.310902108416857

5.3 A_{HU}

- $hu, 1$
 - Value: 4.943755
 - Error: 6.323471336786291
 - Relative error: 0.561226958610952



(a) $A^\beta - (1, 1, 1)$



(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

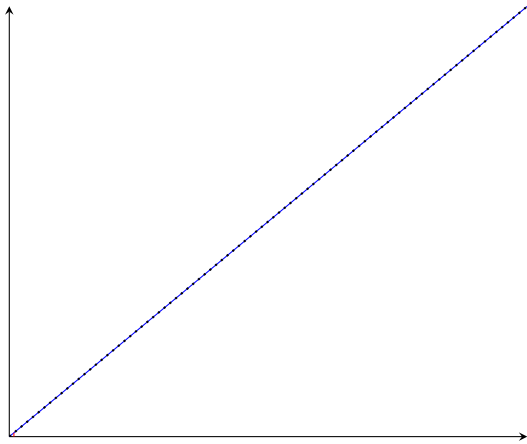


(a) $A^\beta - hu, 1$

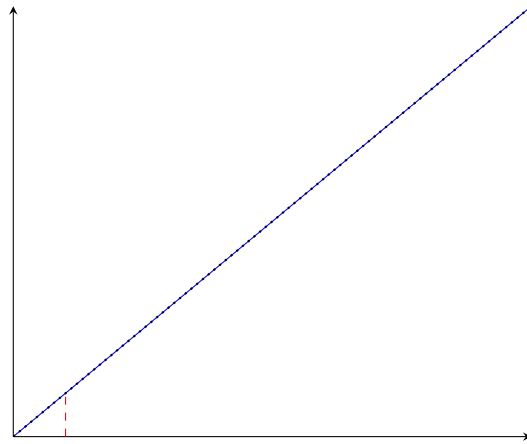
6 A^β Breakpoints

- (1, 1, 1)
 - q : 5.000000
 - $\hat{q}^\beta$: 5.000000
 - q^β : 5.000000
 - Breakpoints: [0, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1, 2, 2)
 - q : 39.365245
 - $\hat{q}^\beta$: 39.365245
 - q^β : 39.365245
 - Breakpoints: [0, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2, 1, 2)
 - q : 28.333333
 - $\hat{q}^\beta$: 28.333333
 - q^β : 28.333333
 - Breakpoints: [0, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 2.981918
 - $\hat{q}^\beta$: 2.981918
 - q^β : 2.981918
 - Breakpoints: [0, 181.61828057849593]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 2$
 - q : 24.973644
 - $\hat{q}^\beta$: 24.973644
 - q^β : 24.973644
 - Breakpoints: [0, 285.4001551947793]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $hu, 1$

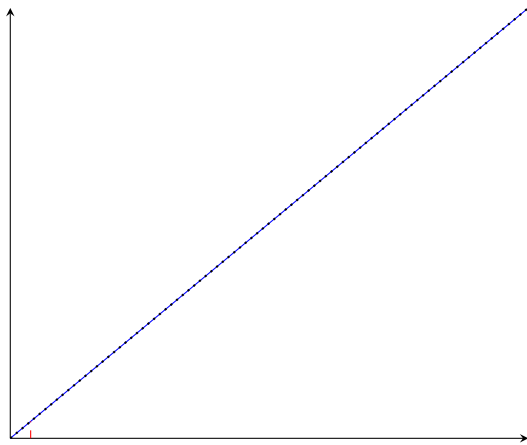
- q : 4.943755
- $\hat{q}^\beta$: 4.943755
- q^β : 4.943755
- Breakpoints: [0, 237.3002543523117]
- Error: 0.0000000000000000
- Relative error: 0.0000000000000000



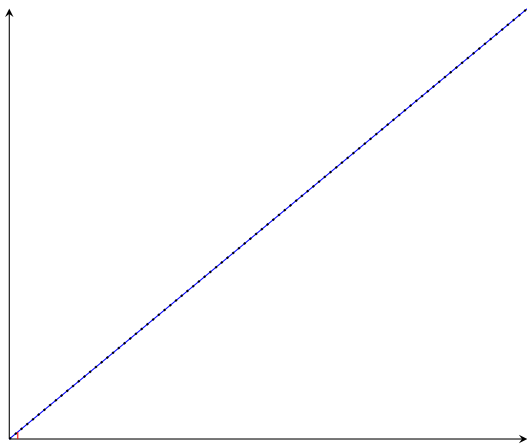
(a) $q^\beta - (1, 1, 1)$



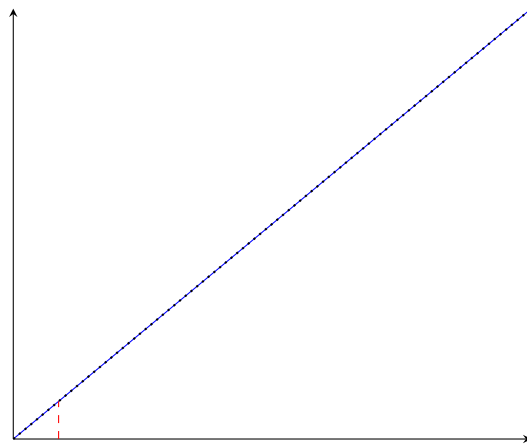
(b) $q^\beta - (1, 2, 2)$



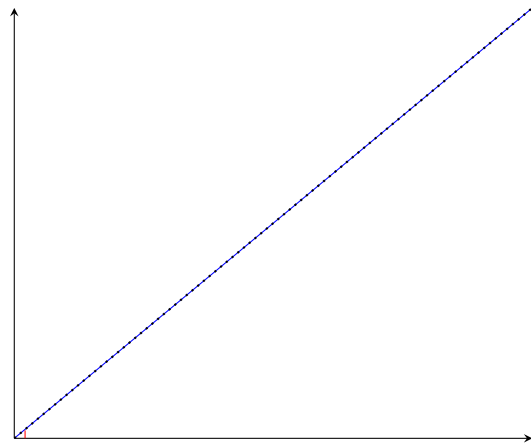
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



(a) $q^\beta - hu, 1$

7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	0.0	$f_{2,2,1}^C$	0.0	$b_{2,2,2}^{H,out}$	4961.363636363635
$z_{1,2,1}$	0.0	$f_{2,2,2}^C$	0.0	$b_{1,1,2}^{C,in}$	1793.75
$z_{2,1,1}$	-0.0	$RecLMTD_{1,1,2}$	0.02282088625994983	$b_{1,2,1}^{C,in}$	6500.0
$z_{2,2,1}$	-0.0	$RecLMTD_{1,2,1}$	0.010552318838737433	$b_{2,1,1}^{C,in}$	7675.0
$z_{2,2,2}$	-0.0	$RecLMTD_{2,1,1}$	0.1	$b_{2,2,1}^{C,in}$	0.0
$z_{hu,2}$	-0.0	$RecLMTD_{2,2,1}$	0.010626071631746416	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.011266071631746425	$b_{1,1,2}^{C,out}$	1793.75
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$b_{1,2,1}^{C,out}$	6500.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.0	$b_{2,1,1}^{C,out}$	7675.0
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.0	$b_{2,2,1}^{C,out}$	0.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.0	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	370.0		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	25.43582274801004	$t_{2,2,2}^H$	462.5		
$dt_{1,2,1}$	150.0	$t_{1,1,2}^C$	410.0		
$dt_{2,1,1}$	24.166666666666668	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	90.0	$t_{2,1,1}^C$	580.0		
$dt_{2,2,2}$	77.93526150792448	$t_{2,2,1}^C$	350.0		
$dt_{2,2,3}$	70.0	$t_{2,2,2}^C$	350.0		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	0.0		
$f_{1,1,2}^H$	0.0	$b_{1,2,1}^{H,in}$	0.0		
$f_{1,2,1}^H$	0.0	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	4961.363636363635		
$f_{2,2,2}^H$	8.409090909090907	$b_{1,1,2}^{H,out}$	0.0		
$f_{1,1,2}^C$	4.374999999999998	$b_{1,2,1}^{H,out}$	0.0		
$f_{1,2,1}^C$	13.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	12.5	$b_{2,2,1}^{H,out}$	0.0		